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ECOMMERCE MANAGEMENT SYSTEM – ERD Diagram

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E commerce management system

1. Introduction

This database is designed for an **E commerce management System** where customers can buy products from sellers. The system manages customers, sellers, products, categories, carts, orders, payments, deliveries, reviews, and tax records.

The goal of this database is to store all information safely and maintain proper relationships between different parts of the system.

2. Entities, Attributes, Relationships, Primary Keys & Foreign Keys

Below are all entities with simple explanations.

2.1 Customer

Attributes:

- C_ID
- Name
- Email
- Phone
- Address, City, State, PinCode
- RegistrationDate

Relationships:

- One customer has one cart
- One customer can place many orders
- One customer can write many reviews

Primary Key:

- C_ID

Foreign Keys:

- None

2.2 Seller

Attributes:

- S_ID
- Name
- Email
- ContactNo
- BusinessName
- Status
- GL_ID

Relationships:

- One seller has many products
- One seller has many tax records

- One seller is linked to one government institution

Primary Key:

- **S_ID**

Foreign Key:

- **GI_ID** → Government Institution

2.3 Government Institution

Attributes:

- GI_ID
- InstitutionName
- Department
- ContactInfo
- LicenseVerificationPortal
- TaxRate

Relationships:

- One institution manages many sellers
- One institution has many tax records

Primary Key:

- **GI_ID**

Foreign Keys:

- None

2.4 Tax Record

Attributes:

- T_ID
- RecordDate
- TaxAmount
- S_ID
- GI_ID

Relationships:

- Each tax record belongs to one seller
- Each tax record belongs to one government institution

Primary Key:

- **T_ID**

Foreign Keys:

- **S_ID** → Seller
- **GI_ID** → Government Institution

2.5 Category

Attributes:

- CategoryID
- CategoryName
- Description

Relationships:

- One category has many products

Primary Key:

- **CategoryID**

Foreign Keys:

- None

2.6 Product**Attributes:**

- P_ID
- Name
- Description
- Price
- StockQuantity
- CategoryID
- S_ID

Relationships:

- One product belongs to one category
- One product belongs to one seller
- One product appears in many cart items
- One product appears in many order details
- One product can have many reviews

Primary Key:

- **P_ID**

Foreign Keys:

- **CategoryID** → Category
- **S_ID** → Seller

2.7 Cart**Attributes:**

- cartID
- C_ID
- CreatedAt

Relationships:

- One cart belongs to one customer
- One cart has many cart items
- One cart is used for one order

Primary Key:

- **cartID**

Foreign Key:

- **C_ID** → Customer

2.8 Cart Item

Attributes:

- Ct_ID
- P_ID
- Quantity
- CartID

Relationships:

- One cart item belongs to one cart
- One cart item belongs to one product

Primary Key:

- **Composite PK:** cartID + P_ID

Foreign Keys:

- **Cart_ID** → Cart
- **P_ID** → Product

2.9 Order**Attributes:**

- O_ID
- OrderDate
- Status
- TotalAmount
- C_ID

Relationships:

- One order belongs to one customer
- One order has many order details
- One order has one payment
- One order has one delivery

Primary Key:

- **O_ID**

Foreign Keys:

- **C_ID** → Customer

2.10 Order Detail**Attributes:**

- OD_ID
- UnitPrice
- Quantity
- Subtotal
- O_ID
- P_ID

Relationships:

- One order detail belongs to one order
- One order detail belongs to one product

Primary Key:

- **OD_ID**

Foreign Keys:

- **O_ID** → Order
- **P_ID** → Product

2.11 Payment**Attributes:**

- P_ID
- PaymentMethod
- PaymentDate
- Amount
- PaymentStatus
- O_ID

Relationships:

- One payment belongs to one order

Primary Key:

- **P_ID**

Foreign Key:

- **O_ID** → Order

2.12 Delivery**Attributes:**

- D_ID
- DeliveryDate
- DeliveryStatus
- TrackingNumber
- O_ID

Relationships:

- One delivery belongs to one order

Primary Key:

- **D_ID**

Foreign Key:

- **O_ID** → Order

2.13 Review**Attributes:**

- R_ID
- ReviewDate
- Rating
- Comment
- C_ID
- P_ID

Relationships:

- One review belongs to one customer

- One review belongs to one product

Primary Key:

- **R_ID**

Foreign Keys:

- **C_ID** → Customer
- **P_ID** → Product

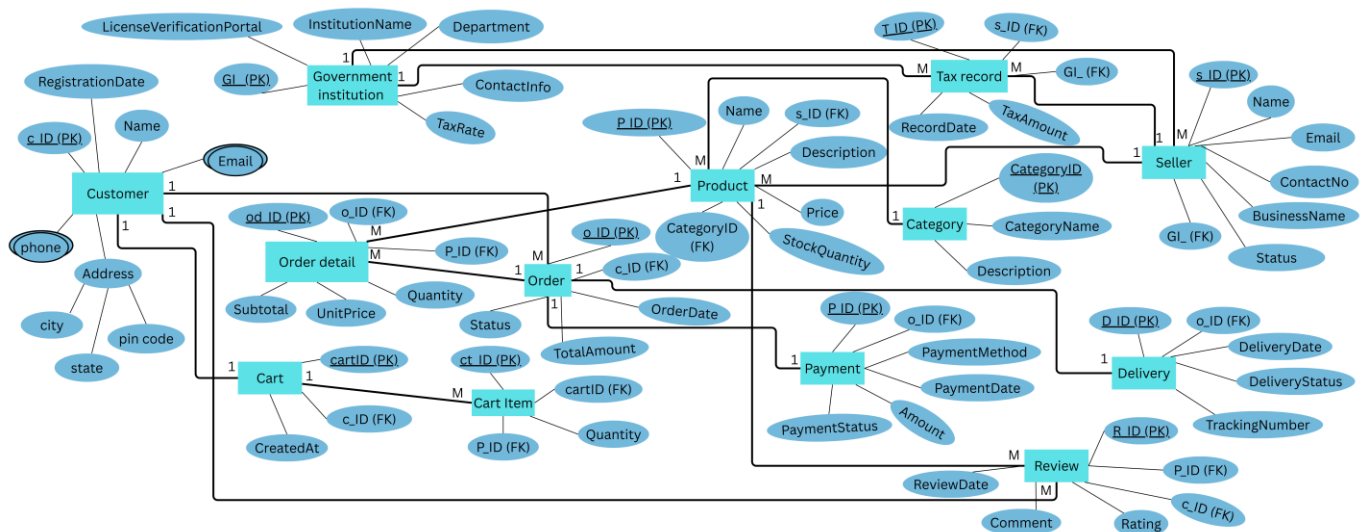
3. Conclusion

This ERD provides a complete database design for a modern e-commerce platform. The system connects customers, sellers, products, orders, payments, deliveries, and reviews in an organized way. It also includes government institutions to monitor taxes, ensuring legal compliance.

The ERD clearly defines all relationships, primary keys, and foreign keys, representing a well-structured relational database model.

This structure supports efficient data storage, retrieval, transaction processing, and secure record management, making it an effective backend system for a real-world e-commerce application.

E commerce management system ERD Diagram :



Github links :

Muhammad Awais

<https://github.com/Awaishkhan0364/IDS-assignment>

Sabtain Ali

<https://github.com/sabtain05/IDS-assignment-.git>

Ahmed Atiq

<https://github.com/Ahmed-create-tech/IDSAssignment-.git>

